

The search continues: Integrating parking into an agent-based transport simulation

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Data-driven agile planning
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MATSim User Meeting 2026

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Motivation

- Parking is an important and widely used policy lever
- Part of supply (mostly) ignored during calibration



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Previous approaches

- **Explicit parking search:**
 - Bischoff and Nagel (2017) model explicit parking search → **computationally too expensive**
 - Tchervenkov (2022) integrated this approach in the MATSim loop, however needed to use a **0.1 % demand sample**
- **A global view to parking behaviour:**
 - Waraich and Axhausen (2012) **parking choice** in MATSim
 - Kohl (2024) parking proxy, extends egress walking times based on spatial-temporal parking demand

Previous approaches



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A parameterization of parking pressure:

- Schlenther et al. (2022), Rybczak (2026) used generalized cost/time penalties
- Parking cost (Rieser and Bischhoff (2024))

→ The parameterization of parking pressure is **computationally efficient**. However, in the existing implementations these are **fixed attributes**

Implementation

Parking occupancy observer:

- Tracks the number of parked vehicles on a link
- At the beginning of each simulation iteration, the initial occupancy is inferred from the locations of vehicles in the selected plans
- During the simulation, occupancy is decreased when a vehicle leaves its parking location and increased when a vehicle completes a parking search and parks

Dynamic parking handler:

- Determines, upon vehicle arrival, whether parking is required for the respective vehicle type (e.g., private cars versus public transport vehicles)

Parking search time calculator:

- When a vehicle reaches its arrival link, the parking search time calculator identifies the surrounding network links using a spatial kernel
- It combines local parking occupancy information with a parking search time function to compute the expected delay

Implementation

Parking search time function:

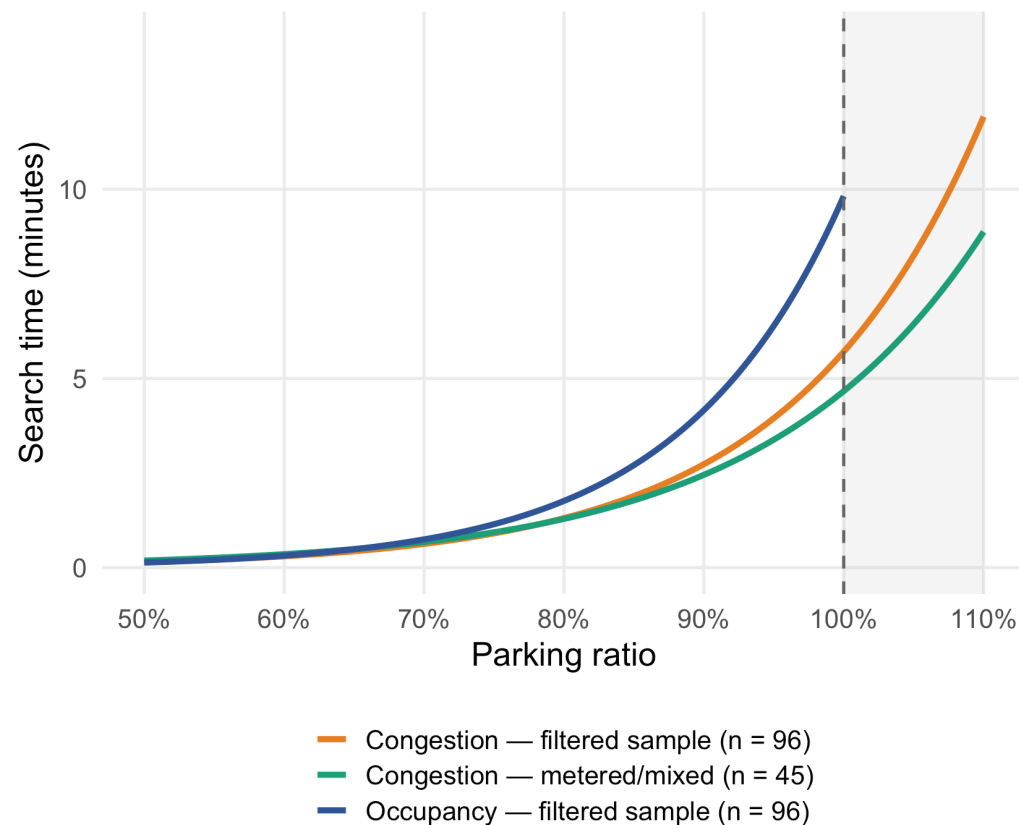
- The search time is calculated using a flexible functional form
- We adopt the formulation of Belloche (2015)

$$t_{\text{parking}} = \alpha \cdot \exp(-\beta \cdot T_c)$$

Congestion ratio, number of parked vehicles divided by the total parking capacity

Empirical Belloche parking-search functions

Predicted parking-search time



Experiments: Toy Example

- Chessboard network
- Fictitious demand 100 agents with 200 trips
- **Normal supply**, 850 spaces in total, and **scarce supply**, 170 spaces in total
- Enlarge chessboard showed, the estimation is computationally efficient and remains responsive to parking scarcity

Model or strategy	Five spaces/link	One space/link	Relative change
Random	1042.52 s	1846.38 s	+77.1%
Benenson	1043.35 s	1832.99 s	+75.7%
Distance memory	1304.60 s	2915.03 s	+123.4%
Parking estimator	348.52 s	523.30 s	+50.1%

Model or strategy	Total runtime [s]	Runtime/iteration [s]	Estimator reduction
Random	121.518	1.215	17.6%
Benenson	128.744	1.287	22.2%
Distance memory	153.308	1.533	34.7%
Parking estimator	100.147	1.001	--

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Experiments in the MATSim Open Berlin Scenario v6.4

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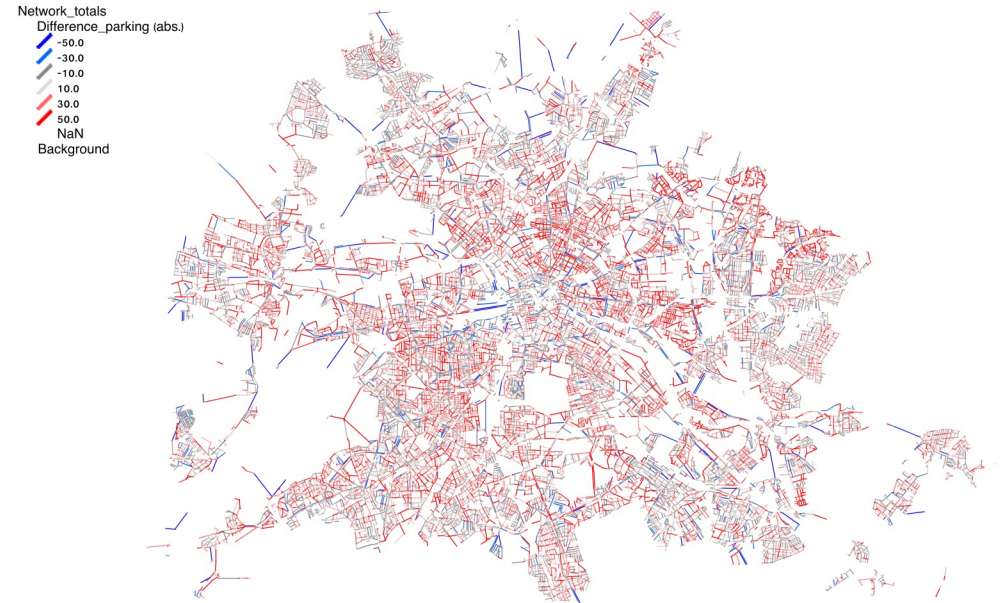
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Supply Berlin

- On street and off-street supply
- On street data set of the city of Berlin
 - Total number of 1.2 million parking spots, with 230,000 located in the city center
 - Assign the regional totals and link-specific assignment to test influence



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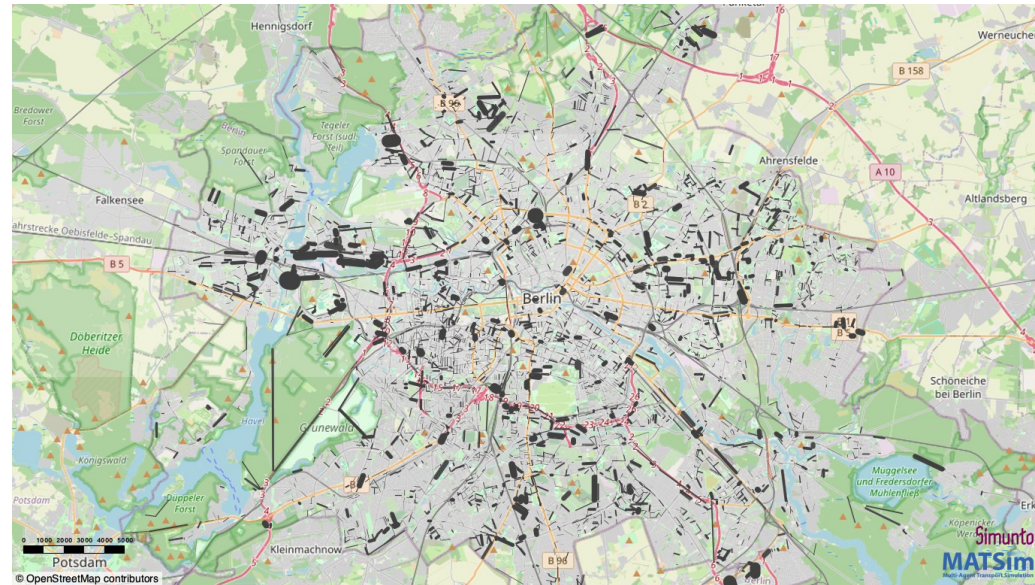
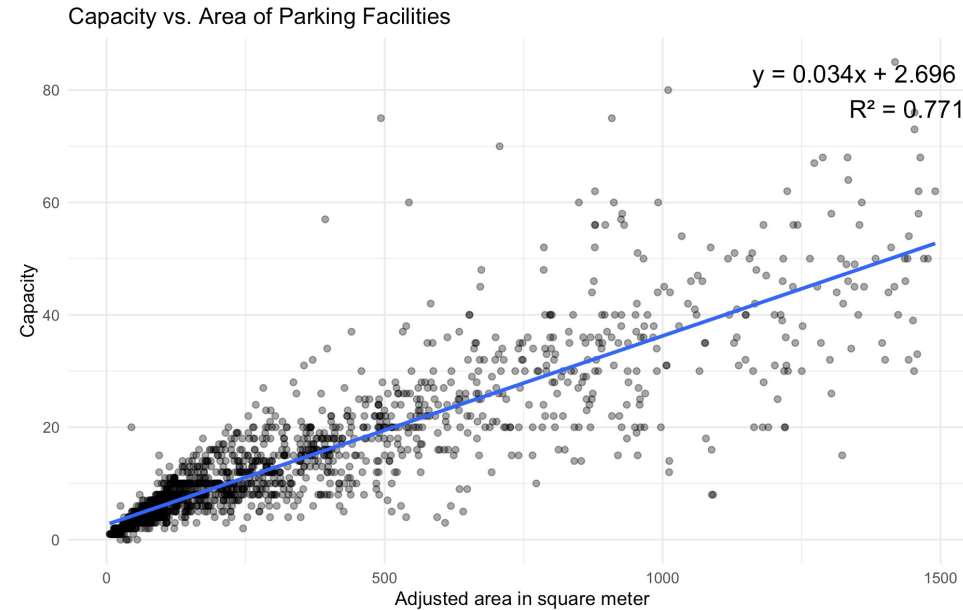
Simunto
MATSim
Multi-Agent Transport Simulation

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Supply Berlin

- Off-street parking is derived from OSM data
- Reported capacities are used directly
- Missing capacities are estimated from area and number of floors
- Each facility is assigned to its nearest eligible network link.



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Experiments without mode choice

- No replanning and different approaches for supply assignment
 - Search times in the Hundekopf are on average higher then in the rest of Berlin
 - Link specific assignments adds heterogeneity in parking search times → use the link-specific parking-data assignment in the subsequent experiments.

Supply assignment	Region	Links	Mean	Median	Std. dev.
Regional-total assignment	Berlin	33,480	0.132	0.0417	2.71
Regional-total assignment	Hundekopf	6,337	0.135	0.0796	0.215
Parking-data assignment	Berlin	33,459	0.328	0.0500	4.75
Parking-data assignment	Hundekopf	6,337	0.521	0.1000	2.37

Mean, Median Std. dev. in minutes

Experiments with mode choice

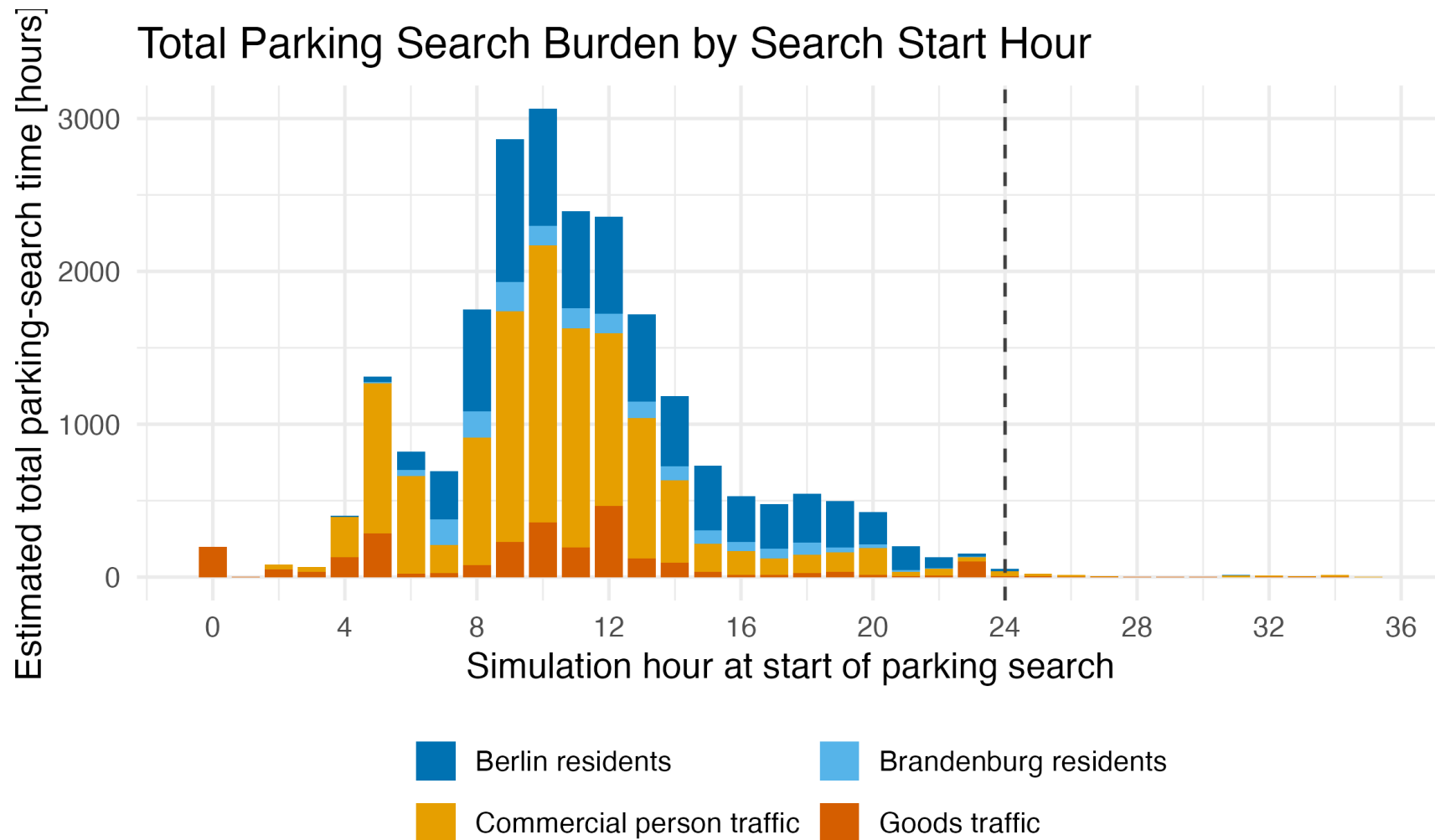
- The number of car trips performed by the residents of Berlin, drops from 2,589,160 to 2,517,080 (-2.78%)
- Parking search times decrease overall

Region	Supply assignment	Links	Mean	Median	SD
Berlin	Parking-data assignment	33,345	0.169	0.039	1.37
Hundekopf	Parking-data assignment	6,308	0.387	0.083	1.56

- Difference in subpopulations

Subpopulation	Persons	Observations	Total [h]	Mean	Median	SD
Commercial passenger traffic	191,670	985,760	11,390	0.693	0.033	14.1
Berlin residents	626,700	2,518,330	7,270	0.173	0.033	1.05
Goods traffic	37,480	222,420	2,560	0.692	0.033	16.1
Brandenburg residents	302,480	1,174,300	1,530	0.078	0.000	1.02

Experiments with mode choice



Dashed line marks simulation hour 24

Experiments with mode choice

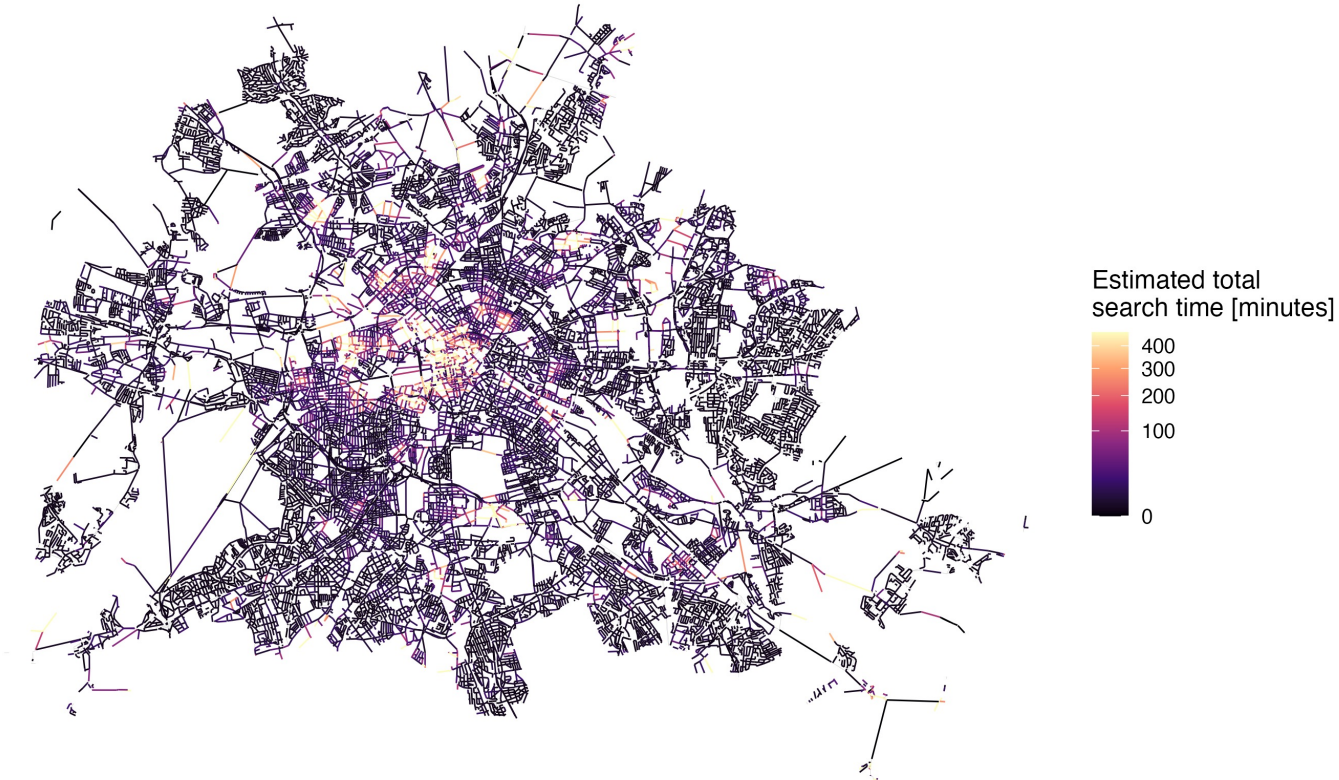
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Total Parking Search Time per Link

Values above the 99th percentile are capped; links without observations are grey



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Validation

- Searched on empirical values for parking search times
 - Mean ranges from 0.60 to 24 minutes
 - Mean values for the same study area differ
 - Values from surveys may structurally differ from studies to GPS data
- Response bias, area, inconsistent definitions
- Commercial agents parking behavior (Kim and Wang (2022))

Table 7: Reported parking search times. Mean, median, and standard deviation are given in minutes. The column n_{stud} denotes the number of extracted not the number of study participants or observed journeys.

Source	Study area	Method	n_{stud}	Mean	Median	SD	Other reported values
Saki and Hagen [2024a]	Germany	GPS/app	1	1.50	0.83		Mean in large-city centres: 1.88 min
Saki and Hagen [2024b]	Germany	GPS/app	1	1.68	0.97	2.37	All journeys; overlapping app dataset
Dalla Chiara et al [2022]	Seattle	GPS	1	1.49	1.15	1.78	With parking information: mean 0.66 min, median 0.37 min, SD 1.47 min
Zhu et al [2020]	Ningbo	Experiment	1	6.03			Range: 1.09-11.55 min
Geotab Inc [2020]	North America	Telematics	4635	4.81	4.58	1.77	Mean: <2 min; cruising rate: 3.6%; peak hour: 4 p.m.
Weinberger et al [2023]	Seattle	GPS, aggregate	1				Mean: <1.5 min; cruising rate: 7.3%; peak hour: 6 p.m.
Weinberger et al [2023]	Atlanta	GPS, aggregate	1				Mean: <1.5 min; cruising rate: 4.9%; peak hour: 6 p.m.
Assemi et al [2023]	Brisbane	Survey					0 min: 25%; <5 min: 40%; >5 min: 35%
Milia et al [2023]	Vesterbro	GPS	1		2.48		
Milia et al [2023]	Frederiksberg	GPS	1		1.98		
Milia et al [2023]	Islands Brygge	GPS	1		1.82		
Weinberger et al [2023]	Chicago	GPS, disaggregate	1				Mean: <3 min; cruising rate: 6.8%; peak hour: 5 p.m.
Mazurkiewicz et al [2021]	Athens	GPS	1	2.18	0.95	3.54	25th percentile: 0.42 min; median: 0.95 min; 75th percentile: 2.33 min
Dalla Chiara and Goodchild [2020]	Seattle	GPS	1	2.30			Commercial vehicles
Weinberger et al [2023]	Washington, DC	GPS, aggregate	1	2.00			Approximate mean: 2 min; cruising rate: 5.8%; peak hour: noon
Qin et al [2020]	Beijing	Survey	1	1.00		2.00	No cruising: 42%; within 1 min: 32%; 1-2 min: 13%; 2-5 min: 9%; >5 min: 4%
Cao et al [2019]	Zurich	Survey and simulation	1				Survey shares and simulation range: 0-14 min
Lee et al [2017]	Brisbane	Survey	1	14.57			On-street: 13.41 min; off-street: 15.72 min

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Table 7 continued from the previous page

Source	Study area	Method	n_{stud}	Mean	Median	SD	Other reported values
Holguin-Veras [2019]	New York City	Truck-driver survey	1	24.00			$n = 16$ search-time responses; minimum: 3 min; maximum: 60 min; mode: 20 min
van Ommen et al [2021]	Melbourne	Analytical	1	1.00			Calculated at 90-95% parking occupancy
Brooke et al [2018]	East Midlands, UK	Survey	1	1.55	1.55		Citation-key year differs from the publication year
Alami et al [2018]	San Francisco	Experiment					Range: 0.50-2.00 min
van der Waerden et al [2013]	Turnhout	GPS	1	1.18			
Kelloch [2015b]	Lyon	Survey	10	4.66	3.94	3.62	Statistics calculated from the ten district-average search times reported in Table 2; range: 0.83-11.10 min; not individual-driver statistics
Saki and Hagen [2024c]	Germany	GPS smartphone application	1	1.42	0.83	1.98	All journeys; $n = 3,550$; includes journeys with immediate parking and zero search time; 5th-95th percentile: 0.00-4.98 min
Saki and Hagen [2024d]	Germany	GPS smartphone application	1	1.73	1.07	2.07	Only journeys with positive parking-search duration; $n = 2,924$; 5th-95th percentile: 0.22-5.53 min
Weinberger et al [2020]	Ann Arbor	GPS	1		2.10		
Weinberger et al [2020]	San Francisco (1)	GPS	1		1.95		
Weinberger et al [2020]	San Francisco (2)	GPS	1		2.80		
Mannini et al [2017]	Rome	GPS/PCD					Range: 0-6 min
Ishimochi et al [2019]	Michigan, US	GPS	1	0.75			30% spent no time cruising; approximately 60% cruised for less than one minute; only the top 20% cruised for more than two minutes
Ucheyronkov [2025]	Zurich	GPS	12	2.94	2.75	0.73	Range: 2.10-4.70 min
Mannini et al [2017]	Zurich	GPS	1				Parking search lasted 4 min in 80% of cases

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Table 7 continued from the previous page

Source	Study area	Method	n_{stud}	Mean	Median	SD	Other reported values
Maurer et al [2023]	Zurich	GPS and survey	1	1.52			
van Ommen et al [2017]	Netherlands	National travel survey	1	0.60			Average across car trips; 30% involved some cruising
Loebken and Pishue [2017]	Frankfurt	Survey	1	10.00			Mean on-street search time; off-street mean: 7.00 min; annual search time: 65 h per driver
Loebken and Pishue [2017]	Berlin	Survey	1	9.00			Mean on-street search time; off-street mean: 6.00 min; annual search time: 62 h per driver



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Experiment with recalibration

- Use automatic ASC calibration to match modal split to reference data (Rakow (2026))

Mode	Base ASC	Calibrated ASC	Change
Walk	0.000	0.000	0.000
Car	−0.488	−0.471	+0.017
Public transport	0.432	0.387	−0.045
Bike	−0.872	−0.878	−0.006
Ride	−1.154	−1.199	−0.045

Values are in utils

Conclusion

- We can capture dynamic and spatially heterogeneous parking pressure at low computational cost
- Parking affects mode choice and therefore needs to be included before scenario calibration → finding correct level of abstraction

Current/future work

- Test Shoup-style parking pricing (derived from Heinrich et al. (2026)) and compare it with Berlin's existing pricing scheme
- Compare parking pricing with road pricing and assess how closely a second-best parking policy approaches the benchmark solution

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Thank you!

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